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ADNOC

ADNOC GROUP PROJECTS AND ENGINEERING

GREEN BUILDINGS GUIDELINES

Guideline

APPROVED BY:

24/11/2021

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EFFECTIVE DATE:

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Abu Dhabi National Oil Company

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GROUP PROJECTS & ENGINEERING / PT&CS DIRECTORATE

CUSTODIAN	Group Projects & Engineering / PT&CS
ADNOC	Specification applicable to ADNOC & ADNOC Group Companies

REVISION HISTORY

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Group Projects & Engineering is the owner of this Specification and responsible for its custody, maintenance and periodic update.

In addition, Group Projects & Engineering is responsible for communication and distribution of any changes to this Specification and its version control.

This specification will be reviewed and updated in case of any changes affecting the activities described in this specification.

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INTER-RELATIONSHIPS AND STAKEHOLDERS

a. The following are inter-relationships for implementation of this Specification:

   i. ADNOC Upstream and ADNOC Downstream Directorates; and

   ii. ADNOC Onshore, ADNOC Offshore, ADNOC Sour Gas, ADNOC Gas Processing. ADNOC LNG, ADNOC Refining, Fertil, Borouge, Al Dhafra Petroleum, Al Yasat

b. The following are stakeholders for the purpose of this Specification:

   i. ADNOC PT&CS Directorate

c. This Specification has been approved by the ADNOC PT&CS is to be implemented by each ADNOC Group company included above subject to and in accordance with their Delegation of Authority and other governance-related processes in order to ensure compliance.

d. Each ADNOC Group company must establish/nominate a Technical Authority responsible for compliance with this Specification.

DEFINITIONS

"ADNOC" means Abu Dhabi National Oil Company.

"ADNOC Group" means ADNOC together with each company in which ADNOC, directly or indirectly, controls fifty percent (50 %) or more of the share capital.

"Approving Authority" means the decision-making body or employee with the required authority to approve Policies and Procedures or any changes to it.

"Business Line Directorates" or **"BLD"** means a directorate of ADNOC which is responsible for one or more Group Companies reporting to, or operating within the same line of business as, such directorate.

"Business Support Directorates and Functions" or **"Non-BLD"** means all the ADNOC functions and the remaining directorates, which are not ADNOC Business Line Directorates.

"CEO" means chief executive officer.

"**Group Company**" means any company within the ADNOC Group other than ADNOC.

"Guideline" means this Green Buildings Guidelines.

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1 GENERAL

1.1 Introduction

Green building or sustainable building is the practice/holistic approach that may be followed in the building design, construction, operation, maintenance & demolition to minimize the building's impact on the environment, the occupants, the community and to increase the efficiency of the building with reduction in the energy, resources, waste & harmful/CO2 emissions.

1.2 Purpose

This document provides general guidelines related to planning, design, operation, construction, replacement, use, occupancy, location and maintenance of buildings that are required to be green building certified as per project requirements. The aspects discussed in the guideline shall be verified during the design / engineering phase and considered as applicable.

Some of the features provided in this guideline may be applied in other buildings that are not targeted for green building certification to enhance sustainability.

1.3 Definitions and Abbreviations

1.3.1 General Terminology

The following defined terms are used throughout this specification:

"**[PSR]"** indicates a mandatory Process Safety Requirement

"**COMPANY**" means ADNOC, ADNOC Group or an ADNOC Group Company, and includes any agent or consultant authorized to act for, and on behalf of the COMPANY.

"**CONTRACTOR**" means the parties that carry out all or part of the design, engineering, procurement, construction, commissioning or management for ADNOC projects. CONTRACTOR includes its approved MANUFACTURER(s), SUPPLIER(s), SUB-SUPPLIER(s) and SUB-CONTRACTOR(s).

"**MANUFACTURER**" means the Original Equipment Manufacturer (OEM) or MANUFACTURER of one or more of the component(s) which make up a sub-assembly or item of equipment assembled by the main SUPPLIER or his nominated SUB-SUPPLIER.

"may" means a permitted option

"shall" indicates mandatory requirements

"should" means a recommendation

"**SUB-CONTRACTOR**" means any party engaged by the CONTRACTOR to undertake any assigned work on their behalf. COMPANY maintains the right to review all proposed SUB-CONTRACTORs; this right does not relieve the CONTRACTOR of their obligations under the Contract, nor does it create any contractual relationship between COMPANY and the SUB-CONTRACTOR.

"SUPPLIER" means the party entering into a Contract with COMPANY to provide the materials, equipment, supporting technical documents and/or drawings, guarantees, warranties and/or agreed services in accordance with the requirements of the purchase order and relevant specification(s). The term SUPPLIER includes any legally appointed successors and/or nominated representatives of the SUPPLIER.

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"**SUB-SUPPLIER**" means the sub-contracted SUPPLIER of equipment sub-components software and/or support services relating to the equipment / package, or part thereof, to be provided by the SUPPLIER. COMPANY maintains the right to review all proposed SUB-SUPPLIERS, but this right does not relieve the SUPPLIER of their obligations under the Contract, nor does it create any contractual relationship between COMPANY and any individual SUB-SUPPLIER.

1.3.2 Technical Terminology

The following definitions/terminologies related to the field of Green buildings to be noted.

Table 1 List of Definitions and Terminologies

Definitions and Terminologies	
ADUPC	Abu Dhabi Urban Planning Council
ASHRAE	American Society of Heating, Refrigeration and Air-conditioning Engineers
EPA	Environmental Protection Agency (US)
ESTIDAMA	This is an Abu Dhabi Government program (Estidama means 'sustainability' in Arabic) coordinated by Abu Dhabi Urban Planning Council and it aims to create guidelines and regulations to ensure sustainable (green) design, operating and maintenance of all types of buildings and communities within the Emirate. Estidama program includes sets of discretionary guidelines and mandatory regulations and adopt "Estidama Pearls" to certify the sustainable buildings and communities. The certification levels for this system are 1 pearl, 2 pearls, 3 pearls, 4 pearls and 5 pearls.
IAQ	Indoor Air Quality
LEED	The Leadership in Energy and Environmental Design (LEED) Green Building Rating System, developed by the USGBC for design, construction, operations and maintenance of high-performance, sustainable buildings. The certification levels for this system are Certified, Silver, Gold and Platinum.
SMACNA	Sheet Metal and Air Conditioning National Contractors Association
SRI	Solar Reflectance Index
USGBC	The U.S.Green Building Council (USGBC), founded in 1993, is a non-profit trade organization that developed Leadership in Energy and Environmental Design (LEED) rating system and promoting the green building industry, including environmentally responsible materials, sustainable architecture techniques and public policy.
VOC	Volatile Organic Compound

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SECTION A - GENERAL

2 REFERENCE DOCUMENTS

2.1 International Codes and Standards

The following Codes and Standards shall form a part of this specification. When an edition date is not indicated for a Code or Standard, the latest edition in force at the time of the contract award shall apply:

2.1.1 Green Building Manuals

UNITED STATES GREEN BUILDING COUNCIL (USGBC)

LEED v4.1 for Building Design & Construction (BD+C)

ABU DHABI URBAN PLANNING COUNCIL (ADUPC)

The Pearl Rating System for Estidama- Building Rating System, version 1.0 - Urban Planning Council, Abu Dhabi.

Pearl Building – Guide for Consultants, Version 1

2.1.2 Others

ADIECC Abu Dhabi International Energy Conservation Code

2.2 ADNOC Specifications

The following ADNOC general specifications, as available with respective Group Company and as applicable to the Project, shall form a part of this guideline. When an edition date is not indicated for a document, the latest edition in force at the time of contract award shall apply.

Concrete Supply and Construction Specification	AGES-SP-01-001
Structural Steel Supply, Fabrication & Erection Specification	AGES-SP-01-002
Structural Design Basis – Onshore Specification	AGES-SP-01-003
Roads and Paving Design and Construction Specification	AGES-SP-01-005
Drainage Systems Design and Construction Specification	AGES-SP-01-007
Waste Management Guidelines	TBC
Architectural Design Basis - Specification	TBC
Architectural Finishes - Specification	TBC
Masonry and Thermal and Moisture Protection	TBC
HVAC Design Basis - Specification	TBC
HVAC Equipment	TBC

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3 DOCUMENT PRECEDENCE

It shall be CONTRACTOR's responsibility to be, or to become, knowledgeable of the requirements of the referenced Codes and Standards.

The CONTRACTOR shall notify the COMPANY of any apparent conflict between this specification, the related data sheets, the Codes and Standards and any other specifications noted herein. Resolution and/or interpretation precedence shall be obtained from the COMPANY in writing before proceeding with the design/manufacture.

In the event of a conflict between documents, the following hierarchy of adherence shall be followed:

a. UAE Statutory Legislation and Regulations

b. ADNOC HSE Standards

c. Purchase Order of Contract documents including Project drawings and Specifications

d. Company General Specifications and Standards

e. International Codes and Standards

4 TECHNICAL DEVIATION CONTROL

Any technical deviations to the Purchase Order and its attachments including, but not limited to, the COMPANY's General Specifications shall be sought by the CONTRACTOR only through technical deviation request format. Technical deviation requests require COMPANY'S review/approval, prior to the proposed technical changes being implemented. Technical changes implemented prior to COMPANY approval are subject to rejection.

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SECTION B – TECHNICAL REQUIREMENTS

5 GENERAL

5.1 Sustainability Impacts of Green Buildings

a. Green buildings improve public health, safety and general welfare by enhancing the design and construction of buildings, through the use of building concepts having a positive environmental impact or a reduced negative impact and encouraging sustainable construction.

b. The following performance enhancements can be obtained by adopting the green building practice:

   i. Conserve natural resources and reduce greenhouse gas emissions

   ii. Improve air and water quality

   iii. Improve public health, safety and general welfare

   iv. Protecting occupant health and improving employee productivity

   v. Reduction in water and energy consumption and waste

   vi. Reduce solid waste, pollution and environmental degradation

   vii. Reduce operating / maintenance / life cycle cost.

5.2 Performance Rating Requirements

a. In general, any of the Green building rating systems (LEED/ESTIDAMA) may be followed with COMPANY approval. However, for buildings outside ADNOC concession area requiring building permit from Local Authority, ESTIDAMA rating system shall be followed.

b. Subject to PROJECT / COMPANY requirements of Green building certification, CONTRACTOR shall engage LEED/ ESTIDAMA Accredited Professional (with demonstrable experience in LEED/ ESTIDAMA) or Consultancy having adequate experience with LEED/ ESTIDAMA requirements, to manage/ coordinate all the activities during DEFINE stage, detailed engineering and construction works so that appropriate LEED/ ESTIDAMA rating can be achieved economically.

c. At the DEFINE stage, CONTRACTOR shall study options for different LEED/ ESTIDAMA ratings, prepare techno-commercial comparison tables by acquiring a qualified Chartered Surveyor to conduct a life cycle cost (LCC) analysis to evaluate, compare various design options and submit the same along with his recommendation, for review / approval of COMPANY.

d. Special attention shall be given to choice of construction type, building materials, maximizing daylight / sunlight use, water recycling, energy conservation / renewable energy production, HVAC system, traffic management and all other LEED/ ESTIDAMA requirements in order to achieve appropriate LEED/ ESTIDAMA rating. CONTRACTOR shall study / develop an energy model for the proposed building(s) using approved dynamic simulation modelling software (Hourly Analysis Program (HAP), HEVACOMP or equal approved) and calculate the baseline building energy consumption and savings.

e. Registration with the relevant authorities like United States Green Building Council (USGBC) USGBC/ Abu Dhabi Urban Planning Council (ADUPC) and establishing project account including the submission of necessary documentations to USGBC/ ADUPC online/ electronically during the basic engineering development stage, shall be carried out by the CONTRACTOR.

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f. CONTRACTOR shall prepare and submit a sustainability report that specify the Green Building features to be implemented in the project for achieving the required rating of LEED/ ESTIDAMA Certification. The report shall also include the detailed methodology with all relevant data for attaining the proposed credit points of each category.

g. CONTRACTOR shall arrange for workshops at suitable intervals during the detailed engineering/ design and construction phases of the project with the participation of COMPANY personnel and Green Building Experts to have a common understanding/ awareness and ensure fulfilling all the requirements of targeted rating.

h. CONTRACTOR shall be responsible for the follow-up of registration with USGBC/ ADUPC, complying with agreed regulations of the rating system, providing valid justifications/ proof for obtaining the credits/ points, preparation of detailed documents/ reports for approval by the relevant authorities and completing all the requirements until the Green Building Certification process is finalized and plaque is obtained/ installed on the building. This includes but not limited to, management plans of waste, storm water and Indoor Air Quality, performing computational fluid dynamic (CFD) simulations showing air permittivity and thermal flow within each building envelope, appointing an independent qualified commissioning agent, reporting to the COMPANY as part of the project design team throughout the project stages.

6 SUSTAINABILITY CREDIT CATEGORIES

a. For the purpose of performance rating credits, the green building aspects are categorized as follows:

   i. Integrated Project Planning and design.

   ii. Energy efficiency.

   iii. Water efficiency and conservation.

   iv. Material conservation and resource efficiency.

   v. Environmental air quality

b. These aspects may be applied to the planning, design, operation, construction, replacement, use and occupancy, location, maintenance, removal and demolition of every building or structure or any appurtenances connected or attached to such building structures. These provisions are not intended to prevent the use of any alternate material, appliance, installation, device, arrangement, method, design or method of construction not specifically prescribed by this guideline.

6.1 Integrated Project Planning and Design

a. Outline planning, design and development methods that include environmentally responsible site selection, assessment of natural systems, building design, building siting and development to protect, restore, and enhance the environmental quality of the site and respect the integrity of adjacent properties.

b. A team should be organized in a way that integrates the contribution of different trades and specialties. This coordination is simply a framework to support the development of holistic, creative, and efficient design solutions and extend throughout the project life cycle from inception to design, construction and commissioning.

c. Develop a building program detailing the project's green building requirements. A building program develops a clear statement of the expectations for the building and the function of the

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entire building and its various rooms and related structures within the budget, schedule, and physical constraints of the project.

d. Preservation and use of available natural resources shall be accomplished through evaluation and careful planning to minimize negative effects on the site and adjacent areas as well as preservation of slopes, management of storm water drainage and erosion controls.

e. Design for grading and excavation to work with the existing topography as this will save both construction and maintenance costs. Also, implement erosion-control strategies that are applicable to the site as exposed soil is susceptible to erosion.

f. During the design stage, opt for a practical design rather than an elaborate one and minimize aspects of road length, building footprint, and the actual ground area required for intended improvements. This can greatly decrease volume of utility installations

g. Include an ecologically knowledgeable landscape architect as an integral member of the design team.

h. Use plants to mitigate climate conditions, such as sand storms protection or as shading areas from the scorching sun.

i. Combine utility corridors when feasible and install them along previously disturbed areas or along new road to minimize land disturbance.

j. UAE regional climate is a hot/dry climate that exhibits significant temperature swings during the day and night. To prevent thermal loss and to reduce internal heat gains, measures like construction of thick walls from materials of high mass (such as masonry) with insulation, insulation above the roof slab, application of an external finish in a light color to reflect solar heat, provision of a barrier to restrict ingress of moisture through the building walls, floor and roof, etc. are needed.

k. Building orientation and shading - locate, orient and shade the building as follows:

    i. Orient buildings to maximize the use of solar energy with the building to be exposed to the sun while at the same time designing access roads, landscaping, and ancillary structures to channel wind toward main buildings for cooling.

    ii. Orient windows and place canopies/ sunshades, porches, and trees to shade windows during the summer.

    iii. Effective window placement to provide more natural light and lessen the need for electric lighting during the day.

    iv. Viewing windows can be designed to be permanently shut while more functional windows can be opened to increase ventilation.

    v. Protect the building from thermal loss, drafts, and degradation of the building envelope caused by wind and wind-driven materials such as dust, sand and leaves with building orientation and landscape features.

l. Consider the following in the Deconstruction and Reuse of Existing Structures:

    i. If feasible, disassemble existing buildings instead of demolishing to allow reuse or recycling of building materials.

    ii. Maintain the maximum elements of existing building structure (including framing, structural floor, roof decking etc.) and envelope (e.g. exterior skin, windows, doors etc.).

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iii. Reuse existing interior non-structural elements (e.g. interior walls, doors, floor coverings, ceiling systems etc.) as much as possible in the new building.

iv. Salvage additional items in good condition such as light fixtures, plumbing fixtures, doors etc. for reuse in this project in an onsite storage area or for salvage in dedicated collection bins for use in other projects.

m. Post construction landscapes designs accomplish as many of the following as possible:

i. Areas disrupted during construction are restored to be consistent with native vegetation species and patterns.

ii. Utilize plant and tree species appropriate for the climate zone region.

n. Prepare a transport plan covering the following aspects:

i. Accessibility of the site for all forms of transport.

ii. Road access and safety, pedestrian access and safety, cycling and parking.

o. Include the necessary technology and infrastructure to increase the use of telecommunication. This should help reduce the need to visit the building.

p. Encourage use of low emission and fuel efficient vehicles, e.g. compressed natural gas (CNG), battery-operated vehicles.

q. Walking and cycling transport should be encouraged, especially during the cooler months. Measures that can be considered include:

i. Provide secure cycle parking at convenient locations close to building entrances, and accessed, where possible, by separated and direct cycle ways.

ii. Design and layout the development to ensure that pedestrians can take the shortest possible route on paths that is pleasant to use.

r. Place a minimum of 50% parking spaces under cover.

s. Design safe and secure parking areas and provide clear and concise signage.

t. Provide good access for delivery vehicles and provide sufficient space for manoeuvring, loading and unloading and provide space for storage of wastes, pallets etc.

u. The site impervious paving must be light coloured, with a Solar Reflectance Index (SRI) of at least 29.

v. Roofing Materials: All roofing systems must comply with the following requirements:

i. Low-Sloped roofing less than or equal to 2:12 slope must have an SRI of at least 82.

ii. Steep-Sloped roofing greater than 2:12 slope must have an SRI of at least 39.

6.2 Energy Efficiency

a. Buildings are responsible for approximately 40% of energy use in most countries. A green building should achieve more than a 15% reduction in energy usage when compared to the baseline performance.

b. Building services such as lighting, ventilation, heating and cooling can either wholly or partially be provided in a passive manner and as such, the energy required to meet these service

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demands is reduced. It has been shown that well designed buildings can reduce the energy consumption substantially.

c. Proper use of daylight can reduce the need for electrical light sources and decrease electricity use with the associated costs and pollution. The day-lighting system should be suitably designed to decrease excessive thermal gains and brightness as a result of direct sunlight, which can cause discomfort and vision impairment. In addition to the occupancy sensors and building management system and for areas have a day light effective contribution to the illumination level during the day hours it is strongly recommended to arrange for:

   i. Proper number of local control switch to switch Off/On the light to the convenient user level.

   ii. Arrange the lighting circuit distribution in a manner that partial switching off the artificial light during the day time, will not lead to entirely dimmed area(s) and/or presence of shady corners / pockets

d. All equipment and appliances provided by the builder shall be labelled with any of the following agencies as applicable:

   i. ESMA (a star rating by Emirates Authority for Standardization and Metrology related to energy efficiency)

   ii. ENERGY STAR (a joint program of the U.S. Environmental Protection Agency and the U.S. Department of Energy designed to identify and promote energy-efficient products and practices).

e. Openings in the building envelope separating conditioned space from unconditioned space needed to accommodate gas, plumbing, electrical lines and other necessary penetrations must be sealed. Annular spaces around pipes, electric cables, conduits, or other openings at exterior walls shall be protected against the passage of rodents by closing such openings with cement mortar, concrete masonry or similar method. Other openings like exhaust fans shall have insulated louvers or covers which close when the fan is off.

f. Design steel framing and cladding support system for maximum energy efficiency. Techniques for avoiding thermal bridging in the envelope include: Punching large holes in the stud web without affecting its structural integrity, Spacing the studs as far as possible while maintaining the structural integrity of the structure, Exterior rigid insulation, and Detailed design of intersections of wall openings and building intersections of floors, walls, and roofs.

g. Construct walls, ceilings, roofs, and floors to have high thermal resistance/insulation. Insulating materials made from recycled materials such as cellulose or mineral wool could be used. However, insulating materials that require chlorofluorocarbons (CFCs) or hydro-chlorofluorocarbons (HCFCs) in their production shall not be used, as these are ozone-depleting compounds.

h. For windows Consider using the currently available techniques:

   i. Reflective films;

   ii. Solar optic lens film applied to acrylic panels;

   iii. Specular blinds or mirror panels;

   iv. Moulded acrylic prismatic glazing or prismatic panels;

   v. Holographic or diffraction-grating glazing;

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vi. Recessed windows

vii. Double glazing, with adequate air space in-between the inner & outer panes.

viii. Sliding windows are usually not very tight and could cause leakage of cooled air, therefore the use of this type of windows shall be avoided. Only fixed or hinged windows shall be used.

i. Building entrances can be protected via vestibules/ airlocks to avoid the loss of cooled air to the exterior. Also, careful detailing, weather-stripping, and sealing of the envelope are required to eliminate sources of convective losses.

j. All the external doors shall be provided with automatic door closer (of heavy duty type), to ensure that the door is kept in a closed position at all the time and consequently minimize the escape of cooled air from the building.

k. On-site renewable energy- As far as possible, use on-site renewable energy sources such as solar, etc. wind and geothermal to generate electrical energy in parallel with normal electrical grid in such a way that the normal power supply from electrical grid shall be designed to feed 100% of electrical loads and electrical power from renewable energy shall feed part of electrical loads whenever it is available , e.g. The energy generated by the Solar PV modules will be fed to the Grid during the daytime, at the same time the buildings will be drawing their electricity requirements from the grid normal power supply.

l. The system capacity of renewable energy source depends on space available and orientation of building and has to be studied case by case.

m. Solar Power – Using of solar power for the purpose of connecting loads such as landscaping, irrigation pumps etc. shall be considered subject to the technical compliance with safety.

n. Roof-mounted Solar hot water heater with storage tank to be used instead of electrical water heater, the system shall have the ability to absorb available solar radiation (energy) and converts it to thermal energy by transferring it directly to water to raise its temperature, the gained energy is stored in storage tank for use at any time of the day.

o. Reduce hot-water system standby losses. Losses from distribution piping and hot-water storage tanks can be more than 30 percent of heating energy input. Tank insulation, anti-convection valves and heat traps, as well as smaller heaters with high recovery rates can be considered to reduce these loss factors.

p. Reduce hot-water service temperatures. Confirm the lowest hot-water temperature needed for the usage or equipment. Lowering the hot-water supply temperature reduces initial heating-energy and system losses. This approach should be limited to a minimum water temperature so as not to allow growth of harmful bacteria in piping.

q. Install hot-water system controls. Appropriate controls optimize energy use.

r. Hot water distribution systems should utilize at least one of the following methods or features in the distribution system:

i. An on-demand hot water recirculation system.

ii. A point of use hot water system.

iii. A centrally located hot water heater to minimize the length of piping between the fixtures and water heater.

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iv. A hot water distribution system is designed to keep all hot water piping runs as short as possible.

s. Lighting and thermal comfort controls.

i. Single-occupant spaces. Provide individual controls that meet energy use requirements. Provide individual task lighting and/or day lighting controls for at least 90% of the building occupants. Provide individual thermal comfort controls for at least 50% of the building occupants.

ii. Multi-occupant spaces - Provide lighting and thermal comfort system controls for all shared multi-occupant spaces, such as classrooms and conference rooms.

t. The following Control Systems for Lighting and Air-Conditioning –shall be used in all buildings, as applicable

i. Motion/occupancy sensors and/or timers and/or dimming system shall be used to control lighting.

ii. A/C shall be controlled using motion sensors and/or timers. AC controls shall have occupancy / non-occupancy mode of operation with suitable pre-defined temperature values for each mode.

iii. CO2 sensor and monitoring in the occupied areas/ rooms to control fresh air intake, with necessary control and equipment.

u. In connection with item 6.2.t. above, the following shall be considered in the lighting/control systems design.

i. Motion sensors shall be provided for controlling automatic switching of main indoor lighting (not including emergency, escape, task or architectural lighting and other locations as per safety/operation requirements) at Non-manned Industrial (Process) Buildings, e.g. Substations, Instrument Equipment Rooms, Operator Shelters etc. and at areas in Non-Industrial (Non-Process) Buildings which are not continuously occupied, e.g. open plan offices areas, stores areas etc.

ii. It is not recommended to use the motion sensors to control lighting in Building main entrance/reception areas, stairways and main access corridors. In these areas lights should be switched manually (applicable to both Industrial and Non-Industrial Buildings)

iii. Motion sensors shall be adequately designed to allow for constant detection of building occupants. Adjustable timers shall also be provided to switch off lights after a pre-selected period of non-detection. Lighting switches can be provided with an “On” function, which overrides the Motion Detector and timer functions until the “Off” function is selected.

iv. The lighting system should have the facility to automatically switch off the lights in response to external brightness and in response to adjusted time period.

v. The external lighting should be controlled to ensure that it is not operational during daylight hours, which will reduce energy wastage.

v. Light fittings shall be Light Emitting Diodes (LEDs) except theatres and other multipurpose facilities in which the lighting represent the main element of their function.

w. Cable voltage drop of main feeders and branch circuits should not exceed 2% from nominal voltage.

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x. Review design criteria for occupant needs, including lighting and electric power, comfort, and occupant density. Consider options to reduce the energy needs of user equipment.

y. Select HVAC equipment that remains efficient over a wide range of load conditions. Size systems to accommodate multiple stages of capacity that can be activated in sequence. Buildings to operate at part-load status most of the time. Considering that the peak system loads occur infrequently and are usually caused by the simultaneous occurrence of multiple factors such as peak occupancy, temperature extremes, and use of equipment. HVAC systems need to respond to various loads in order to achieve the greatest overall efficiency.

z. All electrical motor within the building shall be designed to meet the IEC efficiency class IE3.

aa. As practical as possible, design and operate the facility without mechanical cooling and refrigeration equipment. Where mechanical cooling is used, utilize base building HVAC and refrigeration systems for the refrigeration cycle that minimizes direct impact on ozone depletion and global warming. Select HVAC&R equipment with reduced refrigerant charge and increased equipment life. Maintain equipment to prevent leakage of refrigerant to the atmosphere.

bb. Metering of different facilities and activities will assist in identification of inefficient practices or appliances. Installation of electric meters shall be considered as a mandatory requirement accordingly. Installed meters are to be clearly labelled and provided in easily accessible locations. Metering is also required for all onsite energy generating systems. The metering data to be maintained and shall have the capability to provide hourly, daily, weekly, monthly and annual energy consumption for each end-use and compare consumption for the different time frames. The energy meter shall be installed at various MDB's, SMDB's and DB's as applicable, while exception can be applied to customer unit with typical or insignificant loads. Energy Measurement data should be transferred to Building management system for monitoring, analysis and historical records.

cc. It is mandatory to engage a third party commissioning authority (CxA), who has sufficient design and hands-on experience with building mechanical / electrical systems and is independent of the engineering consultants, construction firms, sub-contractors or equipment suppliers. The CxA shall be involved in all the phases of the project from the project inception until the completion of warranty period, to ensure and verify that the building's energy related systems are designed, selected, procured, installed, calibrated, commissioned and operated as per the project requirements, intended design and enhanced commissioning concept.

dd. Dynamic energy simulation will be carried out for the building for optimizing the energy performance according to the building performance rating method outlined in Appendix G of ANSI/ASHRAE/IESNA Standard 90.1, using the minimum acceptable standards for building fabric, HVAC, service water heating, power, lighting and other equipment.

ee. A minimum 20% performance improvement compared to the baseline building performance shall be demonstrated by the energy simulation model as per the methodology outlined within Appendix G of ASHRAE Standard 90.1.

6.3 Water Efficiency and Conservation

a. Infrastructure capacity: Before designs are commissioned, the available infrastructure capacity for mains potable water supply and waste water discharge shall be verified with the end user. This will identify any restrictions which could potentially affect the development.

b. A prediction of water demand and discharge for the building should be conducted. This should be based on the number of building occupants, standard use patterns, and any peak/ future

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demand forecasts made by end user. Benchmark data should also be incorporated to allow for monitoring and future improvements.

c. Consider the conservation in Construction Water as practical as possible. Design for reduced use of potable water in construction period can be achieved through ensuring that minimum potable water use is embodied into materials selection. Identify site construction activities where potable water may be utilized, reduce the need for these activities or specify alternative grey water sources where possible e.g. for washing or damping down.

d. Water-efficient fixtures and appliances shall be considered / used wherever possible. Technologies include the following:

   i. Flow-controllers and regulators;

   ii. Faucet (tap) aerators;

   iii. Push button operated showers in staff and pool areas;

   iv. Dual flush mechanisms for toilets;

   v. Waterless urinals;

   vi. Proximity devices for toilets and faucets;

   vii. Automatic tap operated by sensor.

e. A schedule of plumbing fixtures and fittings that will reduce the overall use of potable water within the building by 20% shall be provided. Each plumbing fixture and fitting shall meet the 20% reduced flow rate specified in Table below (extracted from LEED guidelines).

Table 2 Fixtures Flow Rates

Fixture Type	Normal Flow-rate	Maximum Flow rate at ≥ 20% Reduction
Showerheads	2.5 gpm @ 80 psi (9.46 lpm @ 5.5 bar)	2 gpm @ 80 psi (7.57 lpm @ 5.5 bar)
Lavatory Faucets Residential	2.2 gpm @ 60 psi (8.33 lpm @ 4.13 bar)	1.5 gpm @ 60 psi (5.68 lpm @ 4.13 bar)
Kitchen Faucets	2.2 gpm @ 60 psi (8.33 lpm @ 4.13 bar)	1.75 gpm @ 60 psi (6.62 lpm @ 4.13 bar)
Gravity tank type Water Closets	1.6 gallons/flush (6.06 litres/ flush)	1.28 gallons/flush(3) (4.85 litres/ flush)
Flushometer Tank Water Closets	1.6 gallons/flush (6.06 litres/ flush)	1.28 gallons/flush (3) (4.85 litres/ flush)
Flushometer Valve Water Closets	1.6 gallons/flush (6.06 litres/ flush)	1.28 gallons/flush (3) (4.85 litres/ flush)
Electromechanical Hydraulic Water Closets	1.6 gallons/flush (6.06 litres/flush)	1.28 gallons/flush (3) (4.85 litres/ flush)
Urinals	1.0 gallons/flush (3.79 litres/ flush)	0.5 gallons/ flush (1.89 litres/ flush)

(1) gpm – US gallons per minute

(2) lpm – litres per minute

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(3) Includes water closets with an effective flush rate of 1.28 US gallons or less when tested per ASME A112.19.2 and ASME A112.19.14.

f. Employ strategies that in aggregate result in 20% less water use than the process water use baseline for the building after meeting the commercial equipment and HVAC performance requirements as listed below. For equipment not addressed in the list below, additional equipment performance requirements may be proposed provided documentation supporting the proposed benchmark or industry standard is submitted.

   i. Clothes Washer: CEE Tier 3A

   ii. Dishwasher with Racks: ENERGY STAR or performance equivalent

   iii. Food Steamer: 2 gallons/hour. Use only boiler-less steamers.

   iv. Pre-Rinse Spray Valves: 1.3 gallons/minute

   v. Kitchen Pot-Washing Sinks: 2.2 gallons/minute

   vi. Cooling Towers: 2.3 gallons/ton-hr. water loss

   vii. Ice machine : ENERGY STAR or performance equivalent

g. Wastewater reduction - Each building shall reduce the generation of wastewater by the following methods:

   i. The installation of water-conserving fixtures (water closets, urinals) meeting the criteria established in above sections.

   ii. Utilizing non-potable water systems, like grey water (untreated household waste water which has not come into contact with toilet waste), and treated wastewater (recycled water) etc.

   iii. New buildings and facilities shall be dual plumbed for potable and non-potable water systems for toilet flushing.

h. Consider Water Reclamation by installation of a grey water collection system from bathtubs, showers, bathroom wash basins, laundry water, etc. and use in landscape irrigation, fire protection, outdoor water features, street cleaning, or for industrial purposes such as cooling water or process water.

i. The following shall be considered for the reduction or elimination of potable water in landscape/irrigation works.

   i. Avoiding indoor and outdoor planting with high irrigation needs.

   ii. Provide water efficient landscape irrigation design that reduces by 50% or eliminate the use of potable water beyond the initial requirements for plant installation and establishment. Methods used are use of recycled water, water treated for irrigation purposes and use of grey water.

   iii. Perform a soil/climate analysis to determine appropriate plant material and design the landscape with native or adapted plants to reduce or eliminate irrigation requirements.

   iv. Where irrigation is required, use high-efficiency equipment and/or climate-based controllers.

   v. Install a low-water consumption irrigation system which does not rely on spray type heads.

   vi. Use a zoned Irrigation system.

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j. Sewage Systems: Sewage shall be treated on site to produce sludge and water. The water can be recycled for non-potable uses, reducing water consumption. Sludge can either be taken off-site for disposal at appropriate facilities, or may be biologically digested at on-site facilities, potentially yielding exportable energy from production of methane.

k. The use of septic tanks / soak away systems to collect the sewage does not meet the water conservation / environmental requirements and shall be avoided.

l. Air conditioning condensate: The condensate from air conditioners, dehumidifiers, and refrigeration units can provide facilities with a steady supply of relatively pure water for many processes. Condensate should not be considered potable because it can contain dissolved contaminants and bacteria. This water however, can be piped into a grey water storage tank for reuse for non-potable applications, e.g. irrigation, flushing.

m. As a part of Water management requirement, the following shall be considered:

   i. Metering of different facilities and activities will also assist in identification of inefficient practices or appliances. Installation of water meters shall be considered as a mandatory requirement accordingly. Installed meters are to be clearly labelled and provided in easily accessible locations.

   ii. Designs for water supply and discharge systems should incorporate leak detection systems to allow determination of location for repair. This should include both indoor and outdoor distribution networks.

   iii. Reduce the potential loss or reduction in quality of potable water by ensuring the availability of automatic shut-offs in public access areas to reduce water wastage, ensuring that all storage tanks are covered and ensuring that all potable, grey water and sewage supplies are segregated from each other to avoid cross contamination.

   iv. Select adequate irrigation method. Irrigation methods such as buried drip feed should be utilized and timed to meet with lowest solar insulation and highest plant uptake during the day, i.e. early in the morning. Above ground spray irrigation methods should not be used.

6.4 Material Conservation and Resource Efficiency

a. Achieve material conservation and resource efficiency through reuse of existing building stock and materials; use of recycled, regional, rapidly renewable and employment of techniques to reduce pollution through recycling of materials and reduction of building pollutants prior to occupancy.

b. Regional materials (meeting the specifications) shall be considered as the first preference. Compared to other products in a given product category, select building materials or products for permanent installation on the project that have been harvested or manufactured in UAE. Regional materials shall make up at least 10%, based on cost of total materials value.

c. Materials containing hazardous components such as asbestos and timber treated with chromated copper arsenate (CCA) in structures subject to frequent and intimate human contact shall not be used. The prohibition of hazardous materials should also be included in the project specification documents during the design stage and followed by the contractor during the construction stage.

d. Consider Bio-based Products as indicated below :

   i. Use only bio-based concrete form-release products.

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ii. Solid Wood Products: All new solid-wood-based materials will be certified as “FSC 100%” by an independent third party in accordance with FSC (Forest Stewardship Council) “Principles and Criteria” and will have received Chain-of-Custody Certification as certified by an accredited certification group such as Smart wood or Scientific Certification Systems (SCS).

iii. Other Wood Products: All other new wood-based materials shall be certified by an independent third party in accordance with “Chain-of-Custody (CoC) Certification requirements established by Forest Stewardship Council (FSC)”.

iv. Preservative-treated lumber with chromated copper arsenate (CCA) treatments is not permitted, and lumber with copper-based treatments (such as ACQ) is permitted only for ground-contact applications.

v. Wood-based materials include but are not limited to the following materials (when made from wood), engineered wood products, or wood-based panel products:

- Rough carpentry
- Miscellaneous carpentry
- Heavy timber construction
- Wood decking
- Particleboard
- Plywood
- Metal-plate-connected wood trusses
- Structural glued-laminated timber
- Finish carpentry
- Architectural woodwork
- Wood panelling
- Wood veneer wall covering
- Wood flooring
- Wood lockers
- Wood cabinets
- Wood doors
- Non-rented temporary construction, including bracing, concrete formwork, pedestrian barriers, and temporary protection

e. For new furniture, do not utilize cushioned office seating, and for lounge seating, do not utilize cushioned seating with brominated flame retardants.

f. Consider reuse of existing, previously occupied buildings, including structure, envelope and other internal elements. Remove elements that pose contamination risk to building occupants and upgrade components that would improve energy and water efficiency such as windows, mechanical systems and plumbing fixtures.

g. Recycled content- The potential for recycling or reusing materials should be explored at all stages of a project. Use materials, equivalent in performance to virgin materials, with post-consumer or

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pre-consumer recycled content value (RCV) for a minimum of 10% of the total value, based on estimated cost of materials in the project. Provide documentation as to the respective values.

h. Cement and concrete.

   i. Use concrete made with the following materials:

      1. Slag cement meeting ASTM C989, Standard Specification for Ground Granulated Blast-Furnace Slag for Use in Concrete and Mortars.

      2. Silica fume meeting ASTM C1240, Standard Specification for Silica Fume Used in Cementitious Mixtures.

   ii. Use the following recycled aggregates for non-structural concrete:

      1. Blast furnace slag as a lightweight aggregate.

      2. Recycled concrete that meets grading requirements of ASTM C33/C33-M, Standard Specification for Concrete Aggregates.

i. Other potential materials for recycling are:

   i. Metals are recyclable if they can be separated by type. Accordingly, steel and aluminium building elements have a high recycling value.

   ii. Most plastics are recyclable, but the current rates of recycling are not high because the wide variety of plastics in use makes them difficult to separate. Some plastics such as pure polyvinyl chloride (PVC) would be recycled from buildings more often if designed for easy removal.

   iii. Glass products are recyclable if separated and uncontaminated.

j. Choose materials proven to be characterized by one or more of the following:

   i. Service life: Select materials for longevity and minimal deterioration under conditions of use.

   ii. Reduced maintenance: Select materials that require little maintenance. For those with surface protection, choose materials that do not require frequent applications of toxic or malodorous finishes.

   iii. Recyclability: Select materials that can be re-used or recycled at the end of their service life in the project.

k. Design and provide appropriate materials & protection system for having a weather-resistant exterior wall and durable foundation envelope, in order to prolong the service life of the building.

l. Moisture control. Employ moisture control measures such as:

   i. Design and maintain landscape irrigation systems to prevent spray on structure/building elements.

   ii. Design exterior entries and openings to prevent water intrusion into buildings, using features such as overhangs and recesses, flashings integrated with a drainage plane, and use non-pervious interior finishes in the vicinity of such openings.

   iii. Install gutter and downspout systems to route water away from the building walls / foundation or connect to landscape drains.

m. Construction waste management plan shall be provided and implemented. It shall comply with the requirements of the local jurisdictions waste management ordinance, if any, and AGES

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standard HSE-EN-ST04 and aim for establishment of green methodology in handling with the waste.. It should include the following as a minimum:

i. Identifies the materials to be diverted from disposal by efficient usage, recycling, reuse on the project, or salvage for future use or sale.

ii. Determines if materials will be sorted on-site or mixed.

iii. Identifies diversion facilities where material collected will be taken.

n. Reduce construction waste -Recycle and/or salvage for reuse a minimum of 30% of the non-hazardous construction and demolition debris.

o. Excavated soil and land clearing debris- 100% of trees, stumps, rocks and associated vegetation and soils resulting primarily from excavation and land clearing shall be reused or recycled.

p. Utilize any on-site existing paving materials that are scheduled for demolition as granulated fill or sub-base material, and include the weight of this material in the calculations for material diverted from landfill disposal.

q. Recycling by building occupants shall be sought about it during design and operation stage. Provide readily accessible dedicated areas that serve the entire building and are identified for the depositing, storage, and collection of non-hazardous materials for recycling, including (at a minimum) paper, corrugated cardboard, glass, plastics and metals. Consider employing cardboard balers, aluminium can crushers, recycling chutes and collection bins at individual workstations to further enhance the recycling program. Take appropriate measures for the safe collection, storage and disposal of two of the following: batteries, mercury-containing lamps, and electronic waste.

r. At the time of final inspection, a proper operation and maintenance manual (the manual shall remain with the building throughout the life-cycle of the structure) which includes all of the following (as a minimum) shall be placed in the building:

i. Operation and maintenance instructions for Equipment and appliances.

ii. Information about water conserving landscape and irrigation design and controllers which conserve water.

iii. Information on required routine maintenance measures, including but not limited to, caulking, painting, grading around building, etc.

6.5 Environmental Air Quality

a. Employ Indoor Air Quality (IAQ) measures as follows:

i. Protect on-site absorbent materials from moisture. Remove and replace any materials with evidence of mold, mildew, or moisture infiltration.

ii. Store odorous and high VOC (volatile organic compound) - emitting materials off-site, without packaging, for a sufficient period to allow odors and VOCs to disperse.

iii. When possible, once materials are on the jobsite, install odorous and high VOC-emitting materials prior to those that are porous or fibrous.

iv. Clean oil and dust from ducts prior to use.

v. Provide additional outdoor air ventilation (mechanically ventilated spaces) to improve indoor air quality.

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vi. In order to achieve acceptable level of Indoor Air Quality, UVC (Ultra Violet C band) emitters need to be installed inside all the Air Handling Units/ Fan Coil Units/Package Units (in front of the cooling coil) to control the growth of molds, bacteria, virus and other harmful micro-organisms. Apart from health aspects, other benefits include reduction in maintenance, reduction in energy use to run the air-conditioning and reduction in pollution caused by the HVAC systems.

vii. After construction ends, with all interior finishes installed, flush-out the building by supplying a total air volume of 14,000 cubic feet of outdoor air per square foot (4,267,140 litres of outdoor air per square meter) of gross floor area while maintaining an internal temperature of at least 60°F 15°C) and no higher than 80°F (27°C) and relative humidity no higher than 60%.

b. Covering of duct openings and protection of mechanical equipment during construction shall be considered. At the time of rough installation, or during storage on the construction site and until final start-up of the HVAC equipment, all duct and other related air distribution component openings shall be covered with tape, plastic, sheet metal or other methods acceptable to reduce the amount of dust or debris which may collect in the system.

c. Minimize and control hazardous particulates/ chemical pollutants entry into buildings and cross-contamination of regularly occupied areas. Install permanent entryway systems measuring at least six feet in the primary direction of travel to capture dirt and particulates at entryways directly connected to the outdoors. This shall consider the following:

i. Qualifying entryways are those that serve as regular entry points for building users.

ii. Acceptable entryway systems include, but are not limited to, permanently installed grates, grilles, or slotted systems that allow cleaning underneath.

iii. Roll-out mats are acceptable only when maintained regularly by janitorial contractors as documented in service contract, or by in-house staff as documented by written policies and procedures.

d. Consider the isolation of pollutant sources. In rooms where activities produce hazardous fumes or chemicals, such as laboratory, garages, janitorial or laundry rooms, and copy or printing rooms, exhaust them and isolate them from their adjacent rooms as follows:

i. Exhaust each space with no air recirculation in accordance with ASHRAE 62.1, to create negative pressure with respect to adjacent spaces with the doors to the room closed.

ii. For each space, provide self-closing doors and deck to deck partitions or a hard ceiling.

iii. Install low-noise, vented range hoods for all cooking appliances and in laboratory or other chemical mixing areas.

e. Healthy ventilation delivery - The general purpose of ventilation in buildings is to provide healthy air for breathing by both diluting the pollutants originating in the building and removing the pollutants from it.

f. Mechanical fans drive mechanical ventilation. Fans can either be installed directly in windows or walls, or installed in air ducts for supplying air into, or exhausting air from, a room. The type of mechanical ventilation used depends on climate.

i. Building Roof Layout indicating the separation distances between outdoor air intakes and any exhausts or discharge points comply with ASHRAE 62.1 or local code whichever is more stringent.

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ii. Increase indoor air ventilation of 15% above the outdoor air ventilation rates.

g. In mechanically ventilated buildings, provide regularly occupied areas of the building with air filtration media for outside and return air prior to occupancy that provides at least a Minimum Efficiency Reporting Value (MERV) of 14. (MERV 14 filters are equivalent to 90% to 95% efficiency, helping to control airborne contaminants. Replacing these filters between construction and occupancy is important to ensure that contaminants from the construction process do not adversely affect occupants.)

h. Select HVAC systems that have no impact on the ozone depletion and global warming. Installations of HVAC, refrigeration, and fire suppression equipment shall comply with following:

   i. Install HVAC, refrigeration and fire suppression equipment that do not contain Chlorofluorocarbons (CFCs.).

   ii. Install HVAC, refrigeration and fire suppression equipment that do not contain Hydro chlorofluorocarbons (HCFCs) and Halons.

   iii. Extruded polystyrene insulation (XPS) and closed-cell spray foam polyurethane insulation shall not be manufactured with hydro chlorofluorocarbon (HCFC) blowing agents.

i. The following shall be considered for increased HVAC Distribution Efficiency:

   i. All medium- and high-pressure ductwork systems shall be pressure-tested in accordance with the current SMACNA standards.

   ii. All ductwork (internal and external) shall be insulated.

   iii. Where possible, all air terminal connections shall be hard-connected with sheet metal ductwork. If flexible ductwork is used, no flexible duct extension shall be more than six feet in length.

   iv. All HVAC equipment shall be isolated from the ductwork system with flexible duct connectors to minimize the transmittance of vibration.

   v. All supply and return air branch ducts shall include the appropriate style of volume damper. Air terminal devices such as grilles, registers, and diffusers shall be balanced at duct branch dampers, not at terminal face.

j. Thermal Comfort & Controls (Thermal Zoning) - A thermal zone is a space or collection of spaces having similar space-conditioning requirements, the same heating and cooling set-point, and is the basic thermal unit (or zone) used in modelling the building. A thermal zone will include one or more spaces. Thermal zones may be grouped together, but systems serving combined zones shall be subject to efficiency and control requirements of the combined zones.

k. For the proposed project, demonstrate that all spaces within the building has been modelled to determine zonal cooling demand and designed to have separately controllable thermal zones, as separate zones and controls for a maximum of 35m2 of open plan space on the perimeter of the building and a maximum of 70m2 of open plan space in the internal areas of the building. Each thermal zone must have thermostats that allow control over air speed or temperature as a minimum.

l. For private enclosed spaces intended for individual use, one thermostatic controller must be provided that allows control over air speed or temperature as a minimum.

m. A capillary break shall be installed for indoor moisture control in compliance with at least one of the following:

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i. A 100 mm thick base consisting of 10 mm or larger clean aggregate shall be provided with a vapour barrier in direct contact with concrete and a concrete mix design which will address bleeding, shrinkage, and curling shall be used. For additional information, see ACI 302.2R-06.

ii. Other equivalent methods approved by the enforcing agency.

n. Moisture content of building materials shall be verified during construction. Building materials with visible signs of water damage shall not be installed. Wall and floor framing shall not be enclosed when the framing members exceed 19% moisture content. Insulation products which are visibly wet or have a high moisture content shall be replaced or allowed to dry prior to enclosure in wall or floor/roof cavities. Wet applied insulation products shall follow the manufacturer's drying recommendations prior to enclosure.

o. Install permanent CO2 monitoring equipment that permits adjustment of ventilation system controls and set points that can be adjusted based on human occupancy.

p. Bathroom exhaust fans shall be adequately provided to minimize the effect on the IAQ (indoor air quality) except when a whole building ventilation system is used, a mechanical exhaust fan shall be provided in each room containing a bathtub, shower, or tub/shower combination, WC and urinal. Exception- Multiple bathrooms using a common multiple port fan.

q. Materials, products, furniture, and HVAC systems need regular maintenance, cleaning, and inspections to ensure that they function as designed and to prevent indoor contaminants from developing in these locations. A detailed IAQ (indoor air quality) plan shall be prepared and implemented.

r. Microbial contamination can have deleterious effects on IAQ. It is more relevant to the design and installation of systems and equipment (including plumbing, and drainage) that have to be properly studied to reduce the possibility of problems arising during operation of a building.

s. Environmental Tobacco Smoke (ETS) Control: Prohibit smoking in buildings or effectively control the ventilation air in smoking rooms. Prohibit smoking in common areas, design building envelope and systems to minimize ETS transfer among dwelling units. Provide dedicated external smoking areas with associated facilities.

t. For Acoustical control - Employ building assemblies and components with Sound Transmission Coefficient (STC) values determined in accordance with ASTM E90 and ASTM E413.

u. Gypsum wall board products to be moisture and mold-resistant. Standard paper-faced gypsum wallboard can be used only in dry climates, where wetting during or after construction is not anticipated. In humid climates, where dampness and condensation are a concern, use only non-paper-faced gypsum wallboard. In wet locations a cementitious wallboard, made of portland or magnesium oxide cement, must be used.

v. Fiberglass Insulation: Fiberglass batt insulation shall contain no formaldehyde-based binders or shall be third-party certified for conformance with Greenguard Gold or Indoor Advantage Gold.

w. Duct Acoustical Insulation: Mechanical sound insulation materials within the duct shall consist of an impervious, non-porous coating that prevents dust from accumulating in the insulating materials.

x. Do not burn rubbish, organic matter, etc. or any material on site. Burning of rubbish, organic material, and other material on-site contributes to air pollution.

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y. Herbicides and Pest Control: Herbicides shall not be permitted, and pest control measures shall utilize EPA-registered bio-pesticides only.

z. Outdoor Air Delivery Monitoring:

   i. All spaces with an occupant density greater than 1 person per 40 square feet must include at least one CO2 monitor located between 3 feet and 6 feet above the finished floor.

   ii. All spaces with occupant density less than 1 person per 40 square feet must include a direct outdoor airflow monitor, capable of measuring the minimum outdoor airflow rate within 15% accuracy.

   iii. Monitoring equipment must be configured to generate a building automation system alarm and a visual or audible alert when CO2 concentrations vary by 10% or more from set point.

aa. Adhesives, Sealants, Paints and Coatings:

   i. Submittals for interior adhesives, sealants, paints and coatings which have a potential impact on indoor air must include product data highlighting VOC content/ emission values and verifying that they do not contain urea-formaldehyde.

   ii. Adhesives and Sealants :

      1. A minimum of 95% (by weight) of all adhesives, adhesive bonding primers, adhesive primers, sealants, sealant primers, or any other primer used on the building interior, which have a VOC (Volatile Organic Compounds) content in excess of the limits specified below (in compliance with ESTIDAMA requirements) shall not be used:

Table 3 Adhesives & Sealants VOC limits

Product Type	Maximum grams of VOC per litre of adhesive or sealant, less water and exempt compounds (g/L)
Indoor Carpet Adhesive	50
Carpet Pad Adhesive	50
Wood flooring Adhesive	100
Rubber flooring Adhesive	60
Sub-floor adhesive	50
Ceramic tile adhesive	65
Cove base adhesive	50
Plasterboard and wall panel adhesive	50
Multipurpose construction adhesive	70
Structural glazing adhesive	100
Architectural sealants	250

      2. The VOC limits of the remaining non-compliant adhesives and sealants must be no more than 50% higher than values listed for each product type.

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3. Interior sealants shall not contain mercury, butyl rubber, neoprene, SBR (styrene butadiene rubber), or nitrile.
4. Adhesives used to apply laminates, whether shop-applied or field-applied, shall contain no urea-formaldehyde.
5. For buildings targeting LEED rating, in addition to VOC content mentioned above, all adhesives and sealants, regardless of where they are used, must be tested according to California Department of Public Health (CDPH) Standard Method V1.2-2017 and comply with VOC emission limits in Table 4-1 of the method. Certificates towards the same shall be supplied.

iii. Paints and Coatings:

1. For a minimum 95% area of interior surfaces, use paints and coatings that comply with limits for VOC content as provided below (in compliance with ESTIDAMA requirements):

Table 4 Paints & Coatings VOC limits

	Product Type	WB = Water Based, SB = Solvent Based	g/L of ready to use product
a	Interior matt walls and ceilings (Gloss <25@60º)	WB	30
		SB	30
b	Interior glossy walls and ceilings (Gloss >25@60º)	WB	100
		SB	100
c	Interior trim and cladding paints for wood and metal	WB	130
		SB	300
d	Interior trim varnishes and wood stains, including opaque wood stains	WB	130
		SB	400
e	Interior minimal build wood stains	WB	130
		SB	700
f	Primers	WB	30
		SB	350
g	Binding Primers	WB	30
		SB	750
h	One-pack performance coatings	WB	140
		SB	500
i	Two-pack reactive performance coatings for specific end use such as floors	WB	140
		SB	500
j	Multi-colored coatings	WB	100
		SB	100
k	Decorative effect coatings	WB	200
		SB	200

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2. The VOC limits of the remaining non-compliant paints and coatings must be no more than 50% higher than the values listed above for each product type.
3. For paints used for exterior walls of mineral substrate, limit of VOC content (g/L of ready to use product) shall be as follows: WB = 40; SB = 430.
4. For buildings targeted for LEED rating, in addition to VOC content limitation as indicated above, VOC emissions evaluations shall be tested according to California Department of Public Health (CDPH) Standard Method v1.2-2017 and comply with the VOC limits in Table 4-1 of the method. Certificates towards the same shall be supplied.

iv. Other requirements of paints and sealants are as follows:

1. Interior field applied varnishes and lacquers are not permitted.
2. Interior paints shall not contain antimicrobial additives (such as fungicides and biocides). (Fungicides and biocides are banned from interior paints because those are needed primarily to extend shelf life. Instead, paint should be used promptly and any remaining paint should be recycled.)
3. Aromatic Compounds: Paints and coatings shall not contain more than 1% (by weight) total aromatic compounds (hydrocarbon compounds containing one or more benzene rings).
4. Restricted Components: Paints and coatings shall not contain any of the following: (The list of banned ingredients closely follows Green Seal's Standard GS-11, with the addition of xylene).

- Acrolein
- Acrylonitrile
- Analine dyes
- Antimony
- Benzene
- Butyl benzyl phthalate
- Cadmium
- Di (2-ethylhexyl) phthalate
- Di-n-butyl phthalate
- Di-n-octyl phthalate
- 1,2-dichlorobenzene
- Diethyl phthalate
- Dimethyl phthalate
- Ethylbenzene
- Formaldehyde
- Hexavalent chromium
- Isophorone
- Lead
- Mercury
- Methyl ethyl ketone
- Methyl isobutyl ketone
- Methylene chloride
- Naphthalene

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- Toluene (methylbenzene)
- 1,1,1-trichloroethane
- Vinyl chloride
- Xylene

bb. Floor coverings:

i. Carpet shall achieve an Environmental Performance Score of 0.0200 as determined through an assessment in the Building for Environmental and Economic Sustainability (BEES) software model.

ii. All carpet systems, including adhesives, must meet or exceed the Carpet and Rug Institute Green Label Plus Indoor Air Quality Test Program.

iii. Carpet cushion shall not contain brominated flame retardants.

iv. Carpet tile applications shall be self-adhering.

v. All resilient floor covering must be certified under the UL Greenguard or FloorScore indoor emissions testing programs.

vi. Engineered wood flooring and bamboo flooring must be certified under the UL Greenguard or FloorScore indoor emissions testing programs.

vii. Composite Wood and Agrifiber Binders: All composite wood, agrifiber products, and wood doors shall contain no added urea-formaldehyde resins.

cc. Systems Furniture and Seating:

i. All systems furniture and seating meet the requirements of one of the following:

1. UL Green guard certification
2. SCS Indoor Advantage certification
3. SCS Indoor Advantage Gold certification
4. ANSI/BIFMA Standard e3-2014e Furniture Sustainability Standard and tested in accordance with ANSI/BIFMA Standard Method M7.1-2011 (R2-2016) and as verified by an independent laboratory

ii. Systems furniture and seating made with coatings or sealants that contain any of the following solvents are not permitted: naptha, benzene, toluene, xylene, hexavalent chromium.

dd. Green Housekeeping:

i. Utilize cleaning products that meet the requirements of the Green Seal GS-37 standard or comply with the requirements and maximum VOC limits of Title 17, California Code of Regulations, Division 3, Chapter 1, Subchapter 8.5, Article 2, Regulation for Reducing VOC Emissions from Consumer Products (latest).

ii. Utilize janitorial paper products and trash bags that meet the minimum percentages of post-consumer recycled content and recovered content requirements of EPA's Comprehensive Procurement Guidelines.

iii. Clean materials which are contaminated prior to placing in collection containers. Deliver materials free of dirt, adhesives, solvents, petroleum contamination, and other substances deleterious to the recycling process.

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7 WATER QUALITY

a. Legionella bacteria which is commonly found in water multiply where temperatures are between 20-45°C and nutrients are available. The bacteria are dormant below 20°C and do not survive above 60°C.

b. Legionnaires disease is a potentially fatal type of pneumonia, contracted by inhaling airborne water droplets containing viable Legionella bacteria. Such droplets can be created, for example, by: hot and cold water outlets; atomizers; wet air conditioning plant; and whirlpool or hydrotherapy baths.

c. The key to preventing Legionnaires disease is to make sure that the maintenance and water management programs are in place to identify hazardous conditions and take steps to minimize the growth and transmission of Legionella and other waterborne pathogens in building water systems. Developing and maintaining a water management program is a multi-step process that requires continuous review, plan and execution.

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SECTION C – OTHER REQUIREMENTS

8 QUALITY PLANS

8.1 General

a. Quality plans shall address all aspects related to local conditions, such as climatic conditions, backup facilities, spare parts, transport possibilities, storage facilities, quarries, local Manufacturer/Suppliers, test facilities (field laboratory).

b. Quality Management Systems shall comply with the applicable requirements of BS EN ISO 9001 "Quality Management Systems- Requirements" and BS EN ISO 9004, "Quality management – Quality of an organization – Guidance to achieve sustained success". Written quality plans and procedures shall be submitted for COMPANY approval.

c. Materials should be obtained from the same source(s) throughout the work (including both trial and production mixes).

8.2 Inspection, Testing and Reporting

a. Inspection, Testing & Reporting shall be as per relevant COMPANY specifications and certifying organization standards as applicable.

8.3 Quality Control and Assurance

a. CONTRACTOR shall be solely responsible for quality control of all materials and workmanship. The quality system shall provide for the planned and systematic control of all quality-related activities performed during design and construction. Implementation of the quality system shall be in accordance with the Project Agreement, CONTRACTOR's approved Quality Manual and Project Specific Quality plan.

b. CONTRACTOR shall ensure that SUB-CONTRACTORs, SUPPLIERs and MANUFACTURERs have in place a QA/QC program which clearly establishes the authority and responsibilities of those responsible for the quality system. Persons performing quality functions shall have sufficient and well-defined authority to enforce quality requirements that they initiate or identify and to recommend and provide solutions for quality problems and thereafter verify the effectiveness of the corrective action.

c. Quality System and Quality Control requirements shall be identified and included in CONTRACTOR's Purchase documentation. Based on these requirements, the SUB-CONTRACTORs, SUPPLIERs and MANUFACTURERs will develop QA/QC programs which shall be submitted to COMPANY and CONTRACTOR for review and concurrence. The SUB-CONTRACTOR's, SUPPLIER's and MANUFACTURER's QA/QC programs shall extend to SUB-SUPPLIERs.

d. CONTRACTOR shall submit certified reports of all tests as soon as the tests are completed satisfactorily.

e. COMPANY reserves the right to inspect or to conduct a quality audit of materials, installation and workmanship standards and shall have unrestricted entry to the construction site and to testing laboratories to witness any or all tests. SUB-CONTRACTORs, SUPPLIERs and

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MANUFACTURERs, within 30 days after award but prior to the pre-inspection meeting, shall provide CONTRACTOR and COMPANY with a copy of its Inspection & Testing Plan for review and inclusion of any mandatory CONTRACTOR and COMPANY witness or hold points.

f. COMPANY may reject improper, inferior, defective, or unsuitable materials, installation and workmanship. All materials and workmanship rejected shall be replaced by CONTRACTOR as directed by the COMPANY.

9 SUB-CONTRACTORS AND SUPPLIERS

a. CONTRACTOR shall assume unit responsibility and overall guarantee for the works.

b. CONTRACTOR shall transmit all relevant Purchase Order documents including specifications to his SUB-CONTRACTORs and SUPPLIERs.

c. It is CONTRACTOR'S responsibility to enforce all Purchase Order and Project Specification requirements on its SUB-CONTRACTORS and SUPPLIERS.

d. CONTRACTOR shall submit all relevant SUB-CONTRACTOR and SUPPLIER engineering data to COMPANY.

e. CONTRACTOR shall obtain and transmit all SUB-CONTRACTOR and SUPPLIER warranties to COMPANY.

f. All drawings and engineering data submitted to COMPANY shall be in the English language, and in metric units.

10 MATERIAL CERTIFICATIONS

a. CONTRACTOR is responsible for the supply of all materials conforming to the relevant codes and standards and certifying organization standards, including the supply of all material/ testing certificates.

11 DOCUMENTATION / MANUFACTURER DATA RECORDS

a. CONTRACTOR shall submit for COMPANY approval the drawings, documentation or information as listed in the individual Material Requisitions and Purchase Orders.

b. CONTRACTOR shall maintain a complete up-to-date set of construction drawings at the jobsite.

c. Mutual agreement on scheduled submittal of drawings and engineering data shall be an integral part of any formal Purchase Order.

d. Comments made by COMPANY on drawing submittals shall not relieve CONTRACTOR of responsibility to meet the requirements of ADNOC General Specifications. Comments by COMPANY shall not be construed as permission to deviate from requirements of the Purchase Order unless specific and mutual agreement is reached and confirmed in writing.

e. Each drawing shall be provided with a title block in the bottom right hand corner incorporating the following information:

   i. Official trade name of the company.

   ii. CONTRACTOR's drawing number.

   iii. Drawing title.

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iv. A symbol or letter indicating the revision number.

f. Revisions to drawings shall be identified with symbols adjacent to the alterations. A brief description of each revision shall be given and, where applicable, the authority and date of the revision shall be listed. The term "Latest Revision" shall not be used.

g. Prior to commencement of construction CONTRACTOR shall submit an “Affidavit of Compliance” that confirm that all materials meet the requirements of this Specification.

h. The following documents shall be submitted to COMPANY prior to the start of construction activities:

i. Inspection and Test Plan (ITP).

ii. Inspection Procedures.

iii. Test Procedure Qualification and Test Acceptance Criteria.

iv. Procedure Qualification Record (PQR).

v. Quality control program, quality plan, quality control procedure, including Quality Control Records (QCR’s), records of quality inspection test reports, and test results.

vi. Material Handling Procedure.

vii. Materials and workmanship specifications.

viii. One set of final drawings.

ix. Maintain a complete set of As Built drawings at the jobsite.

x. Method statements covering all construction activities.

xi. Correction of non-conformances including remediation procedures.

xii. Material Traceability Procedure.

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SECTION D – STANDARD DRAWINGS & DATA SHEETS

12 DATASHEET TEMPLATES

Not applicable.

13 STANDARD DRAWINGS

Not applicable.

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SECTION E - APPENDICES

APPENDIX A1. ESTIDAMA CREDIT POINT LIST

The following attachments are the credit point lists for ratings based on ESTIDAMA Pearl Assessment System (Buildings). This is for guidance only and the latest version of this list in practice as released by ESTIDAMA at the time of project registration/ execution shall be used.

SUMMARY OF CREDIT POINTS FOR THE PEARL BUILDING RATING SYSTEM

IDP	Integrated Development Process	Maximum Credit Points
IDP-R1	Integrated Development Strategy	R
IDP-R2	Tenant Fit-Out Design & Construction Guide	R
IDP-R3	Basic Commissioning	R
IDP-1	Life Cycle Costing	4
IDP-2	Guest Worker Accommodation	2
IDP-3	Construction Environmental Management	2
IDP-4	Building Envelope Verification	1
IDP-5	Re-Commissioning	2
IDP-6	Sustainability Communication	2
TOTAL		**13**

R - Mandatory requirement

NS	Natural Systems	Maximum Credit Points
NS-R1	Natural Systems Assessment	R
NS-R2	Natural Systems Protection	R
NS-R3	Natural Systems Design & Management Strategy	R
NS-1	Reuse of Land	2
NS-2	Remediation of Contaminated Land	2
NS-3	Ecological Enhancement	2
NS-4	Habitat Creation & Restoration	6
TOTAL		**12**

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LBo	Liveable Buildings : Outdoors	Maximum Credit Points
LBo-R1	Plan 2030	R
LBo-R2	Urban Systems Assessment	R
LBo-R3	Outdoor Thermal Comfort Strategy	R
LBo-1	Improved Outdoor Thermal Comfort	2
LBo-2	Pearl Rated Communities	1
LBo-3	Accessible Community Facilities	1
LBo-4	Active Urban Environments	1
LBo-5	Private Outdoor Space	1*
LBo-6	Public Transport	3
LBo-7	Bicycle Facilities	2
LBo-8	Preferred Car Parking Spaces	1
LBo-9	Travel Plan	1
LBo-10	Light Pollution Reduction	1
TOTAL		**14**

*LBo-5 – this credit is only applicable to Multi-Residential

LBi	Liveable Buildings : Indoors	Maximum Credit Points
LBi-R1	Healthy Ventilation Delivery	R
LBi-R2	Smoking Control	R
LBi-R3	Legionella Prevention	R
LBi-1	Ventilation Quality	3*
LBi-2.1	Material Emissions : Adhesives & Sealants	1
LBi-2.2	Material Emissions : Paints & Coatings	1
LBi-2.3	Material Emissions: Carpet & Hard Flooring	1
LBi-2.4	Material Emissions : Ceiling Systems	1
LBi-2.5	Material Emissions : Formaldehyde Reduction	2
LBi-3	Construction Indoor Air Quality Management	2
LBi-4	Car Park Air Quality Management	1
LBi-5.1	Thermal Comfort & Controls : Thermal Zoning	1

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LBi	Liveable Buildings : Indoors	Maximum Credit Points
LBi-5.2	Thermal Comfort & Controls : Occupant Control	2*
LBi-5.3	Thermal Comfort & Controls : Thermal Comfort Modelling	2
LBi-6	High Frequency Lighting	1
LBi-7	Daylight & Glare	2*
LBi-8	Views	1*
LBi-9	Indoor Noise Pollution	1
LBi-10	Safe & Secure Environment	1*
TOTAL		**23**

* LBi-1 – a maximum of 2 credit points are available to Retail

LBi-5.2 – this credit is not applicable to Retail

LBi-7 - this credit is not applicable to Retail

LBi-8 - this credit is not applicable to Retail

LBi-10 – a maximum of 2 credit points are available to Schools

PW	Precious Water	Maximum Credit Points
PW-R1	Minimum Interior Water Use Reduction	R
PW-R2	Exterior Water Monitoring	R
PW-1	Improved Interior Water Use Reduction	15
PW-2.1	Exterior Water Use Reduction: Landscaping	8*
PW-2.2	Exterior Water Use Reduction: Heat Rejection	8
PW-2.3	Exterior Water Use Reduction: Water Features	4
PW-3	Water Monitoring & Leak Detection	4
PW-4	Storm water	4
TOTAL		**43**

* PW-2.1 - a maximum of 10 credit points is available to Schools

RE	Resourceful Energy	Maximum Credit Points
RE-R1	Minimum Energy Performance	R
RE-R2	Energy Monitoring & Reporting	R
RE-R3	Ozone Impacts of Refrigerants & Fire Suppression Systems	R

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RE	Resourceful Energy	Maximum Credit Points
RE-1	Improved Energy Performance	15
RE-2	Cool Building Strategies	6
RE-3	Energy Efficient Appliances	3
RE-4	Vertical Transportation	3
RE-5	Peak Load Reduction	4
RE-6	Renewable Energy	9
RE-7	Global Warming Impacts of Refrigerants & Fire Suppression Systems	4
TOTAL		**44**

SM	Stewarding Materials	Maximum Credit Points
SM-R1	Hazardous Materials Elimination	R
SM-R2	Basic Construction Waste Management	R
SM-R3	Basic Operational Waste Management	R
SM-1	Non-Polluting Materials	3
SM-2	Design for Materials Reduction	1
SM-3	Design for Flexibility & Adaptability	1
SM-4	Design for Disassembly	1
SM-5	Modular Flooring Systems	1
SM-6	Design for Durability	1
SM-7	Building Reuse	2
SM-8	Material Reuse	1
SM-9	Regional Materials	2
SM-10	Recycled Materials	6
SM-11	Rapidly Renewable Materials	1
SM-12	Reused or Certified Timber	2
SM-13	Improved Construction Waste Management	2
SM-14	Improved Operational Waste Management	2
SM-15	Organic Waste Management	2
TOTAL		**28**

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IP	Innovating Practice	Maximum Credit Points
IP-1	Innovative Cultural & Regional Practices	1
IP-2	Innovating Practice	2
TOTAL		**3**

Credit Section	Maximum Credit Points
IDP – Integrated Development Process	13
NS – Natural Systems	12
LB - Livable Buildings	37*
PW – Precious Water	43*
RE – Resourceful Energy	44
SM – Stewarding Materials	28
IP – Innovating Practice	3
TOTAL Credit Points	**177***

* LB : Maximum of 36 credit points available for Offices and 30 credit points for Retail.

PW : Maximum of 45 credit points available for Schools.

Total : Excludes Innovating Practice credit points which are offered as bonus credits.

Pearl Building Rating Levels

Requirement	Pearl Rating Achieved
All mandatory credits	1 Pearl
All mandatory credits + 60 credit points	2 Pearl
All mandatory credits + 85 credit points	3 Pearl
All mandatory credits + 115 credit points	4 Pearl
All mandatory credits + 140 credit points	5 Pearl

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APPENDIX A2. LEED CREDIT POINT LIST

The following attachment can be used as check-list for credits based on LEED Rating System [LEED v4.1 for Building Design & Construction (BD+C) – New Construction]. This is for guidance only and the latest version of the list released at the time of registration / execution shall be used.

Y	?	N			
			Credit	Integrative Process	1

0	0	0	Location and Transportation		16
			Credit	LEED for Neighbourhood Development Location	16
			Credit	Sensitive Land Protection	1
			Credit	High Priority Site	2
			Credit	Surrounding Density and Diverse Uses	5
			Credit	Access to Quality Transit	5
			Credit	Bicycle Facilities	1
			Credit	Reduced Parking Footprint	1
			Credit	Green Vehicles	1

0	0	0	Sustainable Sites		10
Y			Prereq	Construction Activity Pollution Prevention	Required
			Credit	Site Assessment	1
			Credit	Site Development - Protect or Restore Habitat	2
			Credit	Open Space	1
			Credit	Rainwater Management	3
			Credit	Heat Island Reduction	2
			Credit	Light Pollution Reduction	1

0	0	0	Water Efficiency		11
Y			Prereq	Outdoor Water Use Reduction	Required
Y			Prereq	Indoor Water Use Reduction	Required
Y			Prereq	Building-Level Water Metering	Required
			Credit	Outdoor Water Use Reduction	2
			Credit	Indoor Water Use Reduction	6
			Credit	Cooling Tower Water Use	2
			Credit	Water Metering	1

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0	0	0	Energy and Atmosphere		33
Y			Prereq	Fundamental Commissioning and Verification	Required
Y			Prereq	Minimum Energy Performance	Required
Y			Prereq	Building-Level Energy Metering	Required
Y			Prereq	Fundamental Refrigerant Management	Required
			Credit	Enhanced Commissioning	6
			Credit	Optimize Energy Performance	18
			Credit	Advanced Energy Metering	1
			Credit	Demand Response	2
			Credit	Renewable Energy Production	3
			Credit	Enhanced Refrigerant Management	1
			Credit	Green Power and Carbon Offsets	2

0	0	0	Materials and Resources		13
Y			Prereq	Storage and Collection of Recyclables	Required
Y			Prereq	Construction and Demolition Waste Management Planning	Required
			Credit	Building Life-Cycle Impact Reduction	5
			Credit	Building Product Disclosure and Optimization - Environmental Product Declarations	2
			Credit	Building Product Disclosure and Optimization - Sourcing of Raw Materials	2
			Credit	Building Product Disclosure and Optimization - Material Ingredients	2
			Credit	Construction and Demolition Waste Management	2

0	0	0	Indoor Environmental Quality		16
Y			Prereq	Minimum Indoor Air Quality Performance	Required
Y			Prereq	Environmental Tobacco Smoke Control	Required
			Credit	Enhanced Indoor Air Quality Strategies	2
			Credit	Low-Emitting Materials	3
			Credit	Construction Indoor Air Quality Management Plan	1
			Credit	Indoor Air Quality Assessment	2
			Credit	Thermal Comfort	1
			Credit	Interior Lighting	2
			Credit	Daylight	3
			Credit	Quality Views	1
			Credit	Acoustic Performance	1

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0	0	0	Innovation		6
			Credit	Innovation	5
			Credit	LEED Accredited Professional	1

0	0	0	Regional Priority		4
			Credit	Regional Priority: Specific Credit	1
			Credit	Regional Priority: Specific Credit	1
			Credit	Regional Priority: Specific Credit	1
			Credit	Regional Priority: Specific Credit	1

LEED BD+C (New Construction) has four levels of certification, depending on the point thresholds achieved:

LEED Rating Levels

Level	Points Achieved
Certified	40-49
Silver	50-59
Gold	60-79
Platinum	80 points and above

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